

Host Integration Server to Azure Migration Discovery Checklist

Before planning a Host Integration Server (HIS) migration to Azure, a structured discovery pass surfaces the dependencies, protocols, and risks that determine whether the migration goes smoothly — or turns into unplanned downtime. Use this checklist to scope that discovery phase.

Who this is for: IT and enterprise architecture teams running Host Integration Server, SNA gateways, or TN3270/TN5250 terminal connectivity into mainframe or AS/400 (IBM i) systems, evaluating a move to Azure.

Discovery scope & boundaries

Discovery covers all hardware, virtual appliances, communication protocols, security certificates, and downstream application dependencies interfacing with legacy mainframe or AS/400 (IBM i) systems.

In scope	Out of scope
• SNA Gateway & Enterprise Server nodes	• Core mainframe batch job scheduling (unless integrated via API)
• TN3270 / TN5250 emulator sessions	• Peripheral hardware/printers on local branch networks
• Host Integration Server (HIS) BizTalk adapters	• End-user client desktop OS version audits
• Custom MQSeries & CPI-C transaction programs	

Four discovery tracks

Run these concurrently — a typical first pass takes about three weeks, scaled to the size of the environment.

Track	Core assessment activities	Deliverable
1. Network & connectivity	Audit IP routing, TN3270 ports (23/992), SNA DLC links, ExpressRoute circuits, firewall rule bases	Network topology diagram & port matrix
2. Protocol & security	Inspect SSL/TLS cipher suites, RACF/ACF2 credential handshakes, Kerberos/NTLM mappings	Security compliance & encryption report
3. Application interfaces	Identify Transaction Integrator (TI) XML schemas, COM components, .NET client bindings, REST wrappers	API & component inventory catalog

4. Capacity & latency	Measure peak concurrent sessions, packet payload sizes, transaction TPS, round-trip latency	SLA & performance baseline document
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Step-by-step discovery procedure

- 1 Inventory extraction:** deploy discovery scripts to enumerate all active Host Integration Server instances and active connection pools.
- 2 Dependency mapping:** trace inbound and outbound API calls from web front-ends down to legacy CICS/IMS regions using distributed tracing.
- 3 Security & authentication audit:** verify directory integration (Active Directory vs. RACF) and map the service principal names (SPNs) used by integration components.
- 4 Data volume profiling:** analyze daily transaction volumes and peak concurrency to size target Azure VM and App Service tiers accurately.

Risk assessment & mitigation

Risk	Severity	Potential impact	Mitigation
Hardcoded IP addresses in legacy client apps	High	Connection failures post-migration to Azure VNet	Azure Private Link & DNS forwarding rules
Undocumented custom COBOL copybooks	Medium	Transaction Integrator schema generation errors	Reverse-engineer via static code analysis
Exceeded latency threshold on SNA over TCP	High	Timeout exceptions in interactive user sessions	Azure ExpressRoute Local with FastPath

Using your discovery findings

Discovery output should feed a **phased** migration plan, not a single cutover:

- Group dependencies into migration waves by risk and business criticality, not by system age
- Treat any undocumented dependency as a blocker to resolve before scheduling a cutover date
- Size target Azure services directly from the capacity & latency baseline you just captured, not from vendor sizing guides
- Treat any High-severity risk from the matrix above as a go/no-go gate for the pilot wave

Planning a Host Integration Server migration?

We run this discovery phase for teams moving off HIS, and turn it into a phased migration plan — particularly for older HIS environments where the gap in supported protocols and adapters versus current platforms is often bigger than it looks on paper.

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